

likelihood.model: A Generic Framework for Likelihood-Based Inference in R

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Summary

likelihood.model is an R ([R Core Team, 2024](#)) package that provides a generic framework for specifying likelihood models and performing inference in the Fisherian tradition following Royall (1997). The package defines a likelihood_model concept through S3 generics—`loglik()`, `score()`, `hess_loglik()`, and `fim()`—that any model class can implement. When only a log-likelihood is provided, the score and Hessian are computed automatically via Richardson extrapolation ([Gilbert & Varadhan, 2019](#)). Users who supply analytical derivatives gain 10–100x speedups without changing the inference interface.

The package offers three model-building strategies at different levels of generality. `likelihood_name()` wraps any standard R distribution (normal, Weibull, exponential, gamma, etc.) into a likelihood model with automatic support for exact, left-censored, right-censored, and interval-censored observations. `likelihood_contr_model`, an R6 class, handles heterogeneous observation types by dispatching to type-specific log-likelihood contributions and summing them under an independence assumption—a pattern common in reliability engineering and survival analysis where a single data set mixes exact failure times with various forms of censoring. Specialized model classes such as `weibull_uncensored` and `exponential_lifetime` demonstrate how to provide analytical derivatives or closed-form MLEs that bypass numerical optimization entirely.

All models produce `fisher_mle` result objects with a full S3 method suite (`coef`, `vcov`, `confint`, `se`, `aic`, `bic`, `summary`) and Fisherian inference functions: `support()`, `relative_likelihood()`, `likelihood_interval()`, `profile_loglik()`, and `evidence()`. Bootstrap sampling distributions and likelihood ratio tests for nested models are also supported.

Statement of Need

Researchers in survival analysis, reliability engineering, and applied statistics who work with censored or heterogeneous data need a framework that separates *model specification* from *inference machinery*. Existing R tools for maximum likelihood estimation either focus on optimization mechanics (requiring users to manage log-likelihoods, gradients, and Hessians manually) or are tightly coupled to specific model families. likelihood.model fills this gap by defining a minimal interface—implement `loglik()` and inherit everything else—that lets researchers focus on the statistical model while the package handles derivative computation, optimization, bootstrap resampling, and Fisherian inference automatically.

The package is designed as infrastructure for an ecosystem of domain-specific packages. It sits between `algebraic.mle` ([Towell, 2024](#)), which provides MLE result algebra (delta method transformations, marginal distributions), and downstream packages for reliability analysis such as `flexhaz`, `maskedcauses`, and `maskedhaz`, which use the `likelihood_contr_model` class to implement likelihood contributions for series systems with masked failure causes.

State of the Field

Several R packages address maximum likelihood estimation, but each occupies a different niche. The `stats4::mle` function and its extension `bbmle` (Bolker, 2023) provide formula-based MLE fitting with profile likelihood and model comparison, but require users to write the negative log-likelihood as a single function and do not provide a generic interface for score functions, Hessians, or Fisherian inference tools such as likelihood intervals and evidence functions. `maxLik` (Henningsen & Toomet, 2011) offers a unified wrapper around multiple optimization algorithms (including Newton-Raphson) with automatic gradient and Hessian computation, but its focus is on the optimization step rather than on defining a reusable likelihood model concept that downstream packages can extend.

For censored data specifically, the survival package (Therneau, 2024) provides the most widely used infrastructure for survival analysis in R, including Cox proportional hazards and parametric models via `survreg()`, but is oriented toward regression modeling rather than exposing a generic likelihood interface. `flexsurv` (Jackson, 2016) extends this with flexible parametric survival models and user-definable hazard functions but shares the same regression orientation. `fitdistrplus` (Delignette-Muller & Dutang, 2015) fits univariate distributions to censored data via several methods (MLE, moment matching, quantile matching) but does not expose score functions, Fisher information, or Fisherian inference tools.

`likelihood.model` differs from these packages in three ways. First, it defines a composable *likelihood model concept* with S3 generics that downstream packages implement, rather than a monolithic fitting function. Second, its contribution-based architecture (`likelihood_constr_model`) supports heterogeneous observation types—mixed exact, left-censored, right-censored, and interval-censored data within a single model—with dynamic dispatch to type-specific likelihood functions. Third, it provides a complete suite of Fisherian inference tools (support functions, relative likelihood, likelihood intervals, profile likelihood, evidence) as first-class citizens rather than post-hoc additions.

Software Design

The package is organized in three layers. The *core layer* defines the `likelihood_model` concept as a set of S3 generics (`loglik`, `score`, `hess_loglik`, `fim`) with default methods that fall back to numerical differentiation via `numDeriv` (Gilbert & Varadhan, 2019). Any object with `"likelihood_model"` in its class vector inherits a complete inference toolkit: `fit()` for MLE via `optim`, `sampler()` for bootstrap sampling distributions via `boot` (Canty & Ripley, 2024), `lrt()` for likelihood ratio tests, and the Fisherian inference functions. The `fisher_mle` result class inherits from `algebraic.mle::mle`, ensuring compatibility with the `algebraic.mle` interface for delta method transformations (`rmap`) and marginal distributions (`marginal`).

The *model builder layer* provides two ready-to-use implementations. `likelihood_name()` constructs a model from any R distribution name by convention—if `dnorm` and `pnorm` exist, `likelihood_name("norm", ...)` produces a complete likelihood model with censoring support. `likelihood_constr_model` uses an R6 class with a closure-based caching strategy: the S3 wrappers (`loglik(model)`, `score(model)`) return functions that cache the data frame split and resolved dispatchers, eliminating repeated overhead during the hundreds of function evaluations that `optim` performs.

The *example layer* provides reference implementations—`weibull_uncensored` with analytical score and Hessian, and `exponential_lifetime` with a closed-form MLE that bypasses optimization—demonstrating how to specialize the generic interface for performance or analytic convenience.

Research Impact Statement

likelihood.model serves as the inference backbone for an ecosystem of R packages targeting reliability engineering with masked failure data. The flexhaz package uses likelihood.model to define distributions via arbitrary hazard functions; maskedcauses and maskedhaz use likelihood_contr.model to implement MLE for series systems where the component cause of failure may be masked or censored; and compositional.mle composes MLE solvers built on likelihood.model via sequential chaining and parallel racing. All packages in this ecosystem are available on the author's [r-universe](#) and are undergoing CRAN submission. The package has been submitted to CRAN as version 0.9.1, has 322 unit tests with approximately 97% code coverage, and includes five vignettes with worked examples spanning basic MLE, censored data, likelihood contributions for series systems, Fisherian inference, and ecosystem integration.

AI Usage Disclosure

The likelihood.model package was designed and implemented by the author without AI assistance. Claude (Anthropic) was used to help write unit tests. The JOSS paper draft was prepared with AI assistance and reviewed by the author for accuracy and completeness.

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